

3.4.1 The Integral Test

Motivation: Finding the exact sum of a series is, in general, difficult. This is because the formula for the n^{th} partial sum of a series is usually quite complicated. Due to this, we will concentrate in the next few sections on simply determining whether or not a series converges without explicitly finding the sum. We will also discuss some ways to obtain estimates for the sum of convergent series.

One series which does not have a “nice” formula for its n^{th} partial sum is: $\sum_{k=1}^{\infty} \frac{1}{k^2}$. Suppose we want to know whether this series converges or diverges.

Theorem (The Integral Test):

Proof: See the textbook.

Note:

Example 1: Determine (with justification) whether the series $\sum_{k=1}^{\infty} \frac{10}{1+k^2}$ converges or diverges.

Example 2: What is the simplest way to determine whether the series $\sum_{k=1}^{\infty} \frac{k+2}{k+1}$ converges or diverges?